

FSF3827 Assignment 5

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Question 1

Part a)

We know that the ECP algorithm solves linearizations of MILP subproblems and adds invalid points as cutting plane constraints to subsequent subproblems until all constraints are satisfied. This process is guaranteed to converge to the optimal solution for convex MINLP problems because the convex function $g(x, y)$ is always underestimated by its first-order Taylor series approximation:

$$g(x, y) \geq g(x^k, y^k) + \nabla g(x^k, y^k)^T \begin{pmatrix} x - x^k \\ y - y^k \end{pmatrix} \quad \forall (x, y)$$

Thus, introducing the linearized constraint creates a valid outer approximation of the original constraint $g(x, y) \leq 0$. This means that our approximate feasible set $\hat{N}_k$ always contains the true feasible region N , so $N \subseteq \hat{N}_k$ for all k . Consequently, the optimal value z_k obtained from the relaxed subproblem must be a valid lower bound to the true global optimum z^* , meaning $z_k \leq z^*$. To see why this sequence converges to the true optimum as $k \rightarrow \infty$, consider that our solutions $\{(x^k, y^k)\}$ are confined within a bounded search space. If the algorithm runs indefinitely, these points cannot spread out forever and must eventually accumulate near a limit point (x^*, y^*) . We can verify this limit point is feasible by looking at the cutting plane constraint added at step k , which the next solution (x^{k+1}, y^{k+1}) must satisfy:

$$g(x^k, y^k) + \nabla g(x^k, y^k)^T \begin{pmatrix} x^{k+1} - x^k \\ y^{k+1} - y^k \end{pmatrix} \leq 0$$

As the points bunch up near the limit, the distance between consecutive iterations approaches zero. This causes the gradient term to vanish, implying that our limit point satisfies the original constraint:

$$g(x^*, y^*) + \nabla g(x^*, y^*)^T \cdot \mathbf{0} \leq 0 \implies g(x^*, y^*) \leq 0$$

Since we have established that the limit point is feasible and our relaxed objective values z_k provided a lower bound throughout, the solution found is indeed the global optimum:

$$\lim_{k \rightarrow \infty} z_k = z^*$$

Part b)

Fletcher and Leyffer prove convergence by demonstrating that the algorithm never visits the same integer assignment twice, which guarantees finite termination given the assumption that the set of integer variables is finite. Their proof distinguishes between two specific cases to ensure that no cycling occurs.

First, if a chosen integer assignment results in an infeasible NLP subproblem, the algorithm solves a feasibility problem to find the point of minimum constraint violation. The proof demonstrates that the specific cut generated from this feasibility solution strictly excludes that integer assignment from being selected in any future Master Problem.

Second, if an integer assignment results in a feasible subproblem, the Master Problem is updated with an optimality cut and a strict upper bound constraint. Because the optimality cut forces the approximation variable to be greater than or equal to the actual function value, while the upper bound constraint requires it to be strictly lower, it becomes mathematically impossible for the Master Problem to select that feasible integer assignment again.

code

See main.ipynb for the implementation of the ECP and OA algorithm with modifications for mixed-integer variables. The ECP was solved in under 200 iterations while OA did it in 67 iterations

Question 2

The new problem formulation helps alot as it effectively cut out all integer assignments for x_i that would lead to infeasible NLP subproblems. This is because the new constraint $\sum_{i=5}^8 z_i \leq 3$ forces the largest $x_i \leq \sqrt{3} \quad \forall i \implies x_i \in \{0, 1\} \quad \forall i$,

Thus, ECP only needed 14 iterations to converge to the optimal solution, while OA only needing 5.

Question 3

a)

Given the equation:

$$\begin{aligned} & \text{minimize} && (x_1 + x_2)^2 - x_2x_3 - 0.25x_4^2 + x_4x_5 \\ & \text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\ & && x_1 + x_5 \geq 2, \\ & && x_1 + x_4 \geq 1, \\ & && x_2 + 0.25x_3 \leq 2.5, \\ & && 0 \leq x_i \leq 3, \quad i = 1, \dots, 5. \end{aligned}$$

this can be expanded to the following equivalent form:

$$\begin{aligned} & \text{minimize} && x_1^2 + 2x_1x_2 + x_2^2 - x_2x_3 - 0.25x_4^2 + x_4x_5 \\ & \text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\ & && x_1 + x_5 \geq 2, \\ & && x_1 + x_4 \geq 1, \\ & && x_2 + 0.25x_3 \leq 2.5, \\ & && 0 \leq x_i \leq 3, \quad i = 1, \dots, 5. \end{aligned}$$

then lets extract and isolate the non-convex terms:

$$\begin{aligned} & \text{minimize} && x_1^2 + 2y_1 + x_2^2 + y_2 + 0.25z + y_3 \\ & \text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\ & && x_1 + x_5 \geq 2, \\ & && x_1 + x_4 \geq 1, \\ & && x_2 + 0.25x_3 \leq 2.5, \\ & && y_1 = x_1x_2, \\ & && y_2 = -x_2x_3, \\ & && z = -x_4^2, \\ & && y_3 = x_4x_5, \\ & && 0 \leq x_i \leq 3, \quad i = 1, \dots, 5. \end{aligned}$$

and now to transform this problem to a convex relaxation, lets use McCormick envelopes to relax / linearize the bilinear terms and we can relax the univariate concave $z = -x_4^2$ with its secant lines passing

its endpoints yielding a convex relaxation of the original problem:

$$\begin{aligned}
& \text{minimize} && x_1^2 + 2y_1 + x_2^2 + y_2 + 0.25z + y_3 \\
& \text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\
& && x_1 + x_5 \geq 2, \\
& && x_1 + x_4 \geq 1, \\
& && x_2 + 0.25x_3 \leq 2.5, \\
& && y_1 \geq 0, \\
& && y_1 \geq 3x_1 + 3x_2 - 9, \\
& && y_1 \leq 3x_1, \\
& && y_1 \leq 3x_2, \\
& && y_2 \leq 0, \\
& && y_2 \leq -3x_2 - 3x_3 + 9, \\
& && y_2 \geq -3x_2, \\
& && y_2 \geq -3x_3, \\
& && z \geq -3x_4, \\
& && y_3 \geq 0, \\
& && y_3 \geq 3x_4 + 3x_5 - 9, \\
& && y_3 \leq 3x_4, \\
& && y_3 \leq 3x_5, \\
& && 0 \leq x_i \leq 3, \quad i = 1, \dots, 5.
\end{aligned}$$

we can now substitute z for $-3x_4$ back into the objective function to obtain the final convex relaxation:

$$\begin{aligned}
& \text{minimize} && x_1^2 + 2y_1 + x_2^2 + y_2 - 0.75x_4 + y_3 \\
& \text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\
& && x_1 + x_5 \geq 2, \\
& && x_1 + x_4 \geq 1, \\
& && x_2 + 0.25x_3 \leq 2.5, \\
& && y_1 \geq 0, \\
& && y_1 \geq 3x_1 + 3x_2 - 9, \\
& && y_1 \leq 3x_1, \\
& && y_1 \leq 3x_2, \\
& && y_2 \leq 0, \\
& && y_2 \leq -3x_2 - 3x_3 + 9, \\
& && y_2 \geq -3x_2, \\
& && y_2 \geq -3x_3, \\
& && y_3 \geq 0, \\
& && y_3 \geq 3x_4 + 3x_5 - 9, \\
& && y_3 \leq 3x_4, \\
& && y_3 \leq 3x_5, \\
& && 0 \leq x_i \leq 3, \quad i = 1, \dots, 5.
\end{aligned}$$

Solving the above convex relaxation yields a lower bound of -3.14 for the original non-convex problem.

b)

Essentially, the RLT concept allows us to generate a tight linear relaxation by moving beyond simple variable bounds and incorporating the problem's actual constraints into the relaxation process. By multiplying existing constraints together or with variable bounds, we create new valid inequalities that better capture the feasible region's shape. This leads to a more accurate approximation of the original non-convex problem, resulting in a tighter lower bound when we solve the relaxation. In contrast, McCormick envelopes only use variable bounds to linearize bilinear terms, which can leave significant gaps between the relaxation and the true feasible region. Thus, RLT leverages the problem's structure to enhance the relaxation quality beyond what McCormick envelopes alone can achieve.